

Tracking EEG Band Power Through General Anesthesia in R: Slow Waves Rise Gradually and Before the Recorded Loss of Consciousness



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Abstract:

Anesthesia is one of the most common pre-surgery procedures that renders a patient unconscious in order to minimize pain during surgery. While a patient is under, brain activity is recorded using an electroencephalogram (EEG). This study aims to identify changes in the power of the four frequency bands (delta, theta, alpha, beta) that the EEG is recorded in. A real EEG recording was analyzed using R by calculating the band power of each of the frequency bands and comparing through three windows: conscious, under, and after waking. After the drug was administered, slow delta waves grew sevenfold, theta by 2.4 times, alpha 1.25 times, and beta fell to 0.54 its original value. Under anesthesia, fast, small waves are gradually replaced by slower, wider waves as brain activity slows down. Using real data allowed us to see that the change happened gradually, not all at once at the loss of consciousness (LOC), and that the patient's outward behavior is not the same as the brain's electrical activity.

Background Information:

- Anesthesia is a very common procedure used during surgeries to put patients to sleep and minimize pain.
- While the patient is under, the brain's electrical signals can be measured using EEG, or electroencephalogram, data.
- The 10-20 system is the international map used to position electrodes on the scalp, named this way due to the 10% and 20% spacings used (Fig. 1). Fz = frontal, middle line.
- Brain activity is typically measured in 4 frequency bands: delta (1 to 4 Hz), theta (4 to 8 Hz), alpha (8 to 13 Hz), and beta (13 to 30 Hz). The proportions of these frequencies change with brain activity.

Research Question:

How does the power of each frequency band change when a patient goes under general anesthesia, and does the change line up with the clinical team's recordings?

The 10-20 System

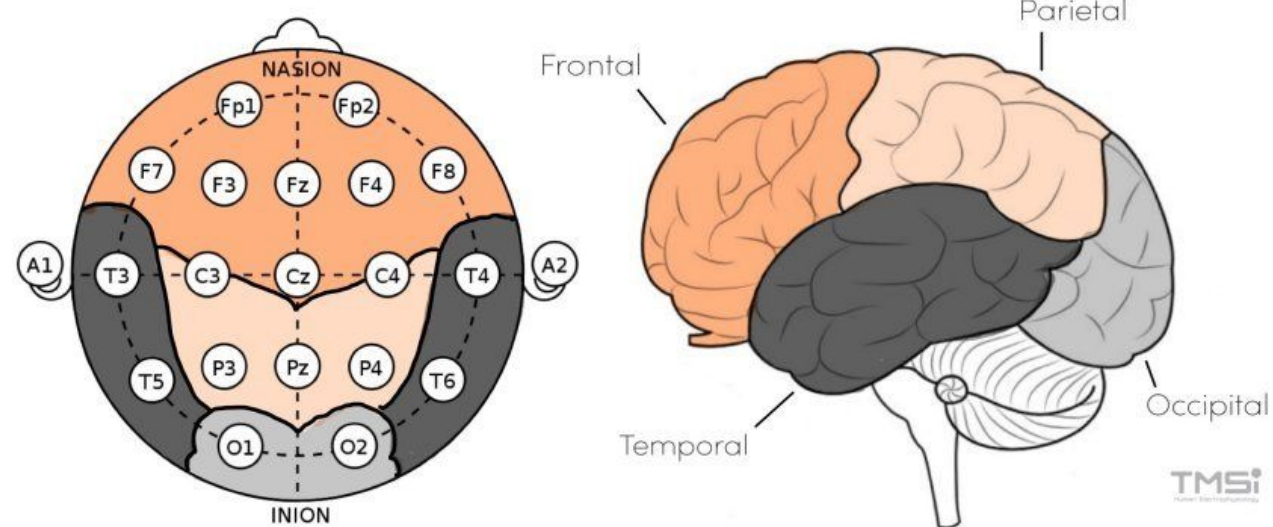


Figure 1. Diagram of the 10-20 system used in most EEG recordings to position electrodes on the scalp.

Method/Materials:

- OpenNeuro dataset ds004541, patient sub_02, session 1
- The **Fz electrode**, the frontal midline electrode, was chosen since it shows the clearest signal with less eye and muscle noise.
 - Fz channel extracted from 60 to 2100 seconds and reduced from 1000 to 250 samples per second
- RStudio (v.4.6.1)**
 - R package “edfReader” was used to open and read the data.
 - R's spec.pgram() function was used for linear detrending and to apply a 10% taper.
 - Raw Fz electrode wanders by about 4800 μ V from non-brain related factors (sweat, body heat, slight electrode movements)
 - Sliding 4 second window, step size = 1
 - Conscious:** 700-880 seconds
 - Under:** 910-1090 seconds
 - After waking:** 2000-2090 seconds
- Loss of consciousness (LOC) and return of consciousness (ROC) were taken from the dataset's events list written by the operating team.
- Power is reported in periodogram units rather than μ V², so only unit-free ratios are interpreted.

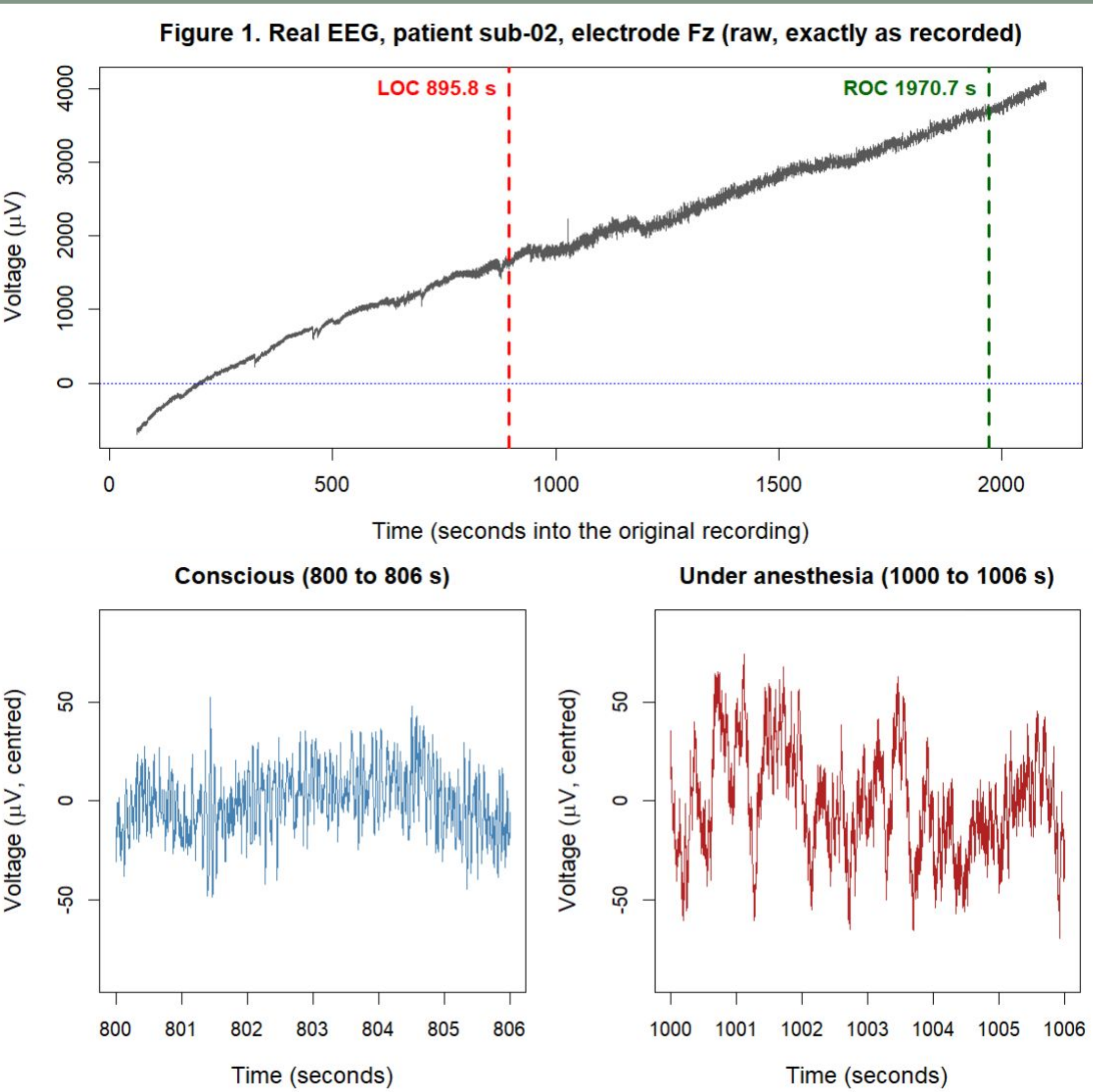


Figure 2. (a) Raw EEG signal exactly as recorded at the Fz electrode. The voltage wanders beyond the expected range due to the warming of the patient's body, sweat, and slight shifting of the electrode. The steady incline seen in the graph has nothing to do with brain activity. (b) Drift is removed by detrending the data. Left panel: 6 seconds of conscious brain signals. Right panel: 6 seconds of unconscious brain signals. The left panel has much tighter signals, staying between plus or minus 50 μ V, while the right panel shows wider signals, swinging out to about 70 μ V.

Results:

- The raw EEG signal wanders by about 4800 μ V due to rising body temperature, sweat and slight shifting of electrode (Fig. 2a).
- Consciousness is characterized by faster, tighter signals while the anesthetized brain shows much wider waves (Fig. 2b)
- Delta band power grew to 6.9 times its original value between 3 minutes before loss of consciousness and 3 minutes after (Fig. 4).
 - Instead of falling, alpha rose by 1.25 of its original value and a new peak appears at 9-10 Hz (Fig. 3).
 - This peak is the result of the coherent frontal alpha oscillations caused by propofol-induced unconsciousness (Purdon *et.al*, 2013).
 - Theta grew by a factor of 2.4 while beta fell by 0.54.
- The delta-to-alpha ratio rose from 0.8 to 4.3 between 3 minutes before and after loss of consciousness.

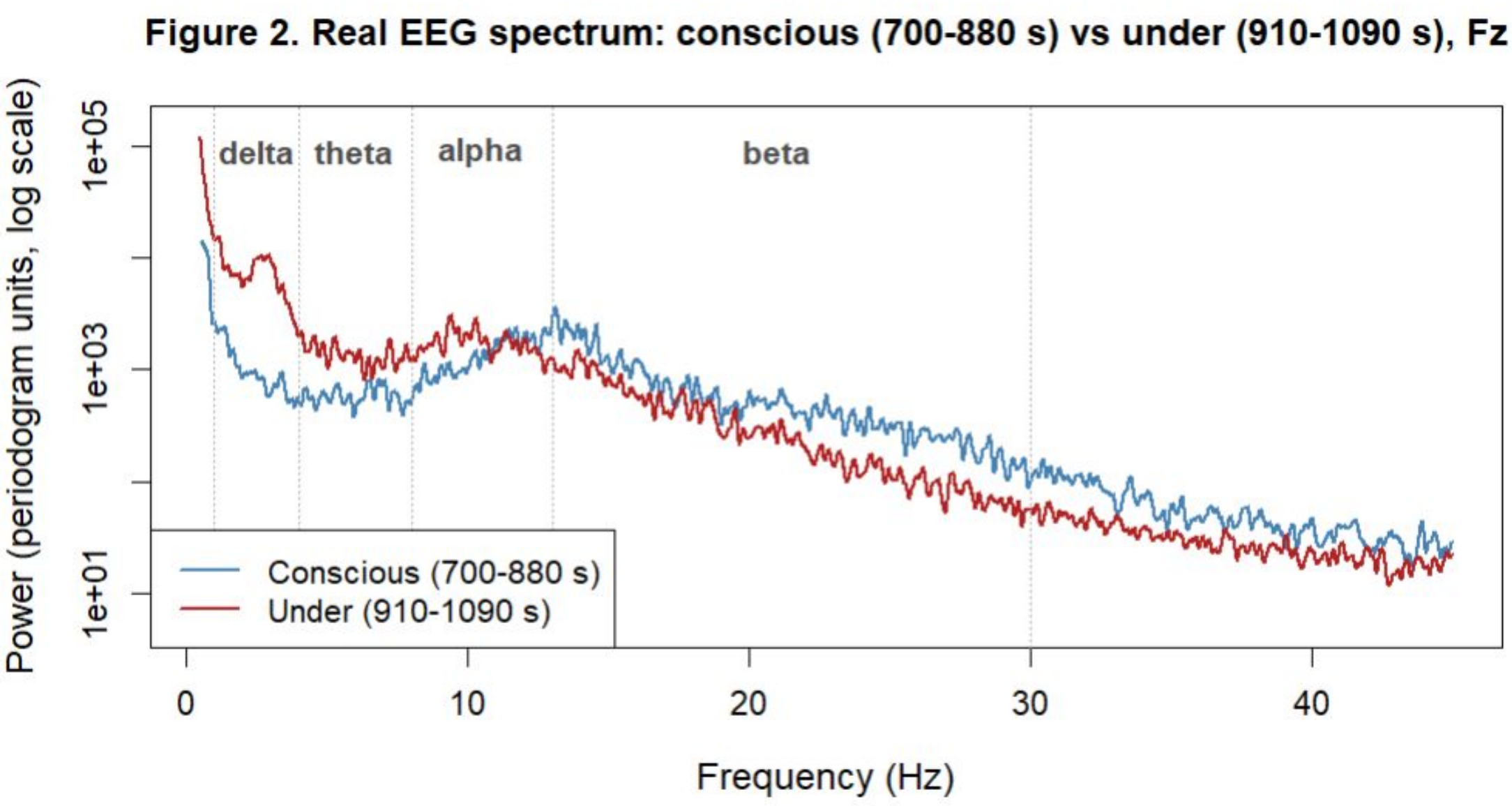


Figure 3. EEG power spectrum at electrode Fz comparing a conscious window (700-880 s) and an anesthetized window (910-1090 s). Slower frequencies (delta and theta) display higher band power under anesthesia than during consciousness. As frequency increases, the conscious line overtakes the under line as fast and alert activity gets quieter with the drug. Alpha power did not fall under anesthesia, instead, there is a distinct bump at 9 to 10 Hz. Power is shown on a log scale in periodogram units.

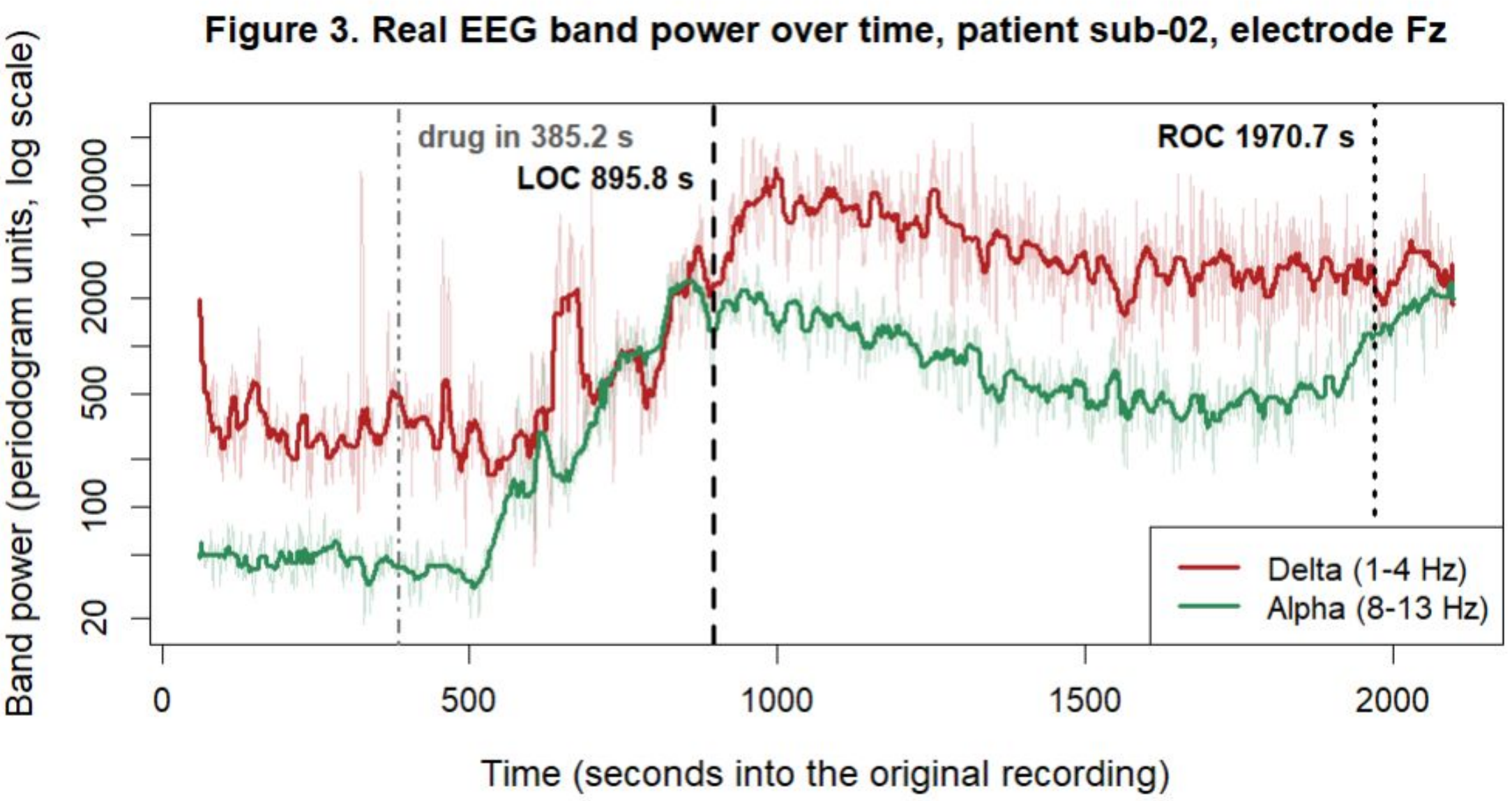


Figure 4. Delta and alpha band power over time at electrode Fz. Drug administration, loss of consciousness (LOC) and return of consciousness (ROC) are marked. Before the drug is administered, delta is already about 6 times greater than alpha. Both bands climb sharply after drug administration, peaking around the LOC and steadily decreasing throughout the anesthetized period. Throughout the recording, delta mainly stays above alpha. After waking, delta and alpha levels don't immediately drop back. Faint lines show raw measurements and thick lines show the smoothed running median.

Discussion:

- Under anesthesia, brain activity slows down greatly, with rapid beta waves gradually getting replaced by slower delta waves – consistent with Hagihira (2015). Alpha was an exception, rising instead.
 - Instead of getting quieter, brain activity got more organized, and started pulsing in a more synchronized manner (Purdon *et.al*, 2013)
 - The change is large enough that it can be measured with a single EEG channel. This is why depth-of-anesthesia monitors can work with just a few electrodes on the scalp.
- The change began within 2 minutes after the drug was administered and was mostly finished before the patient was judged unconscious. LOC is a judgement of behavior, which does not line up with the electrical changes in real data.

Conclusion:

- Delta power rose by a factor of 6.9 from 3 minutes before LOC to 3 minutes after.
 - This change happened gradually and before the recorded LOC.
- Limitations:**
 - This is a case study as it only analyzes a single channel from a single patient, so it is difficult to generalize the findings.
 - Although the data was detrended, movement, muscle, and blinking were not removed. Also, slow drift could have contributed to delta band power.
 - This is an observational study, so it can't be concluded that the correlation between delta power and time was caused by the drug.
- Future work:**
 - Comparison between all 8 patients in the dataset to confirm the same pattern in each.
 - Compare changes in band power between patients who woke up with and without complications to see whether EEG data can be used to predict after effects.

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References:

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